

CAR PRICE PREDICTION

Exploratory Data Analysis Report

Dataset: car_price_prediction.csv

2,500 Records | 10 Features | 7 Brands | 28 Models | 4 Fuel Types

18 Visualisations | Python / Pandas / Seaborn / Matplotlib

Project Overview & Dataset

Comprehensive Exploratory Data Analysis of the Car Price Prediction dataset — 2,500 automobile records across seven global brands, four fuel types, and two transmission categories, spanning 18 visualisations covering pricing patterns, brand positioning, mileage behaviour, and feature inter-relationships.

Dataset Summary

File	car_price_prediction.csv
Total Records	2,500 rows
Total Features	10 columns
File Format	CSV (Comma-Separated Values)
Domain	Automotive / Market Pricing
Primary Task	Regression / Price Prediction
Missing Values	None
Duplicate Records	None detected

Feature Reference

Car ID (Integer)	Unique identifier for each vehicle record (1-2,500)
Brand (String)	Manufacturer - Audi, BMW, Ford, Honda, Mercedes, Tesla, Toyota
Model (String)	Specific vehicle model - 28 unique models
Year (Integer)	Manufacturing year - range: 2000 to 2023
Engine Size (Float)	Engine displacement in litres - range: 1.0L to 6.0L
Fuel Type (String)	Petrol, Diesel, Electric, or Hybrid
Transmission (String)	Manual or Automatic
Mileage (Integer)	Total distance driven in miles - range: 15 to 299,967
Condition (String)	New, Used, or Like New
Price (Float)	Market price in USD - range: \$5,011 to \$99,983

Statistical Summary

Year	Min: 2000 Mean: 2011.6 Median: 2012 Max: 2023 Std: 6.99
Engine Size	Min: 1.0L Mean: 3.47L Median: 3.4L Max: 6.0L Std: 1.43
Mileage	Min: 15 Mean: 149,750 Median: 149,085 Max: 299,967 Std: 87,920
Price	Min: \$5,011 Mean: \$52,638 Median: \$53,485 Max: \$99,983 Std: \$27,296

Data Cleaning & Preprocessing

Duplicate Removal	df.drop_duplicates() - no duplicates detected
Null Check	df.isnull().sum() - zero missing values confirmed across all 10 columns
Unique Value Counts	value_counts() run on Brand, Fuel Type, Transmission, Condition
Correlation Matrix	df.corr(numeric_only=True) - all off-diagonal values near zero
Outlier Detection	IQR method applied to Price - Q1/Q3 bounds computed
Skewness & Kurtosis	scipy.stats.skew / kurtosis computed on Price column

Price Distribution

The overall price distribution establishes the baseline pricing landscape, revealing how vehicle prices spread across the full range from budget to premium segments.

Code

```
sns.histplot(df['Price'], kde=True, color='skyblue')
```

Visualisation



Figure 1: Price Distribution with KDE

Key Insights

1	Price distribution appears fairly uniform across all ranges, with no single dominant price band driving the overall shape.
2	A slight increase in density is observed in the mid-to-high price segments, suggesting a marginally higher concentration of listings in that range.
3	The distribution shows no pronounced skewness in either direction — prices are relatively evenly spread across the full range.
4	The absence of sharp peaks indicates diverse pricing with no strong clustering around specific price points.
5	The smooth KDE curve confirms the absence of extreme outliers with disproportionate influence on the overall distribution.
6	Both lower and upper price ranges record slightly fewer observations compared to the middle band, reinforcing the broadly uniform shape.
7	Overall, the distribution reflects a balanced mix of budget, mid-range, and premium vehicles indicating strong segment diversity.

Price Distribution by Brand

Brand-level boxplots reveal how pricing spreads within each manufacturer, separating premium positioning from mass-market patterns.

Code

```
sns.boxplot(x='Brand', y='Price', hue='Brand', data=df, palette='Set2',
legend=False)
```

Visualisation

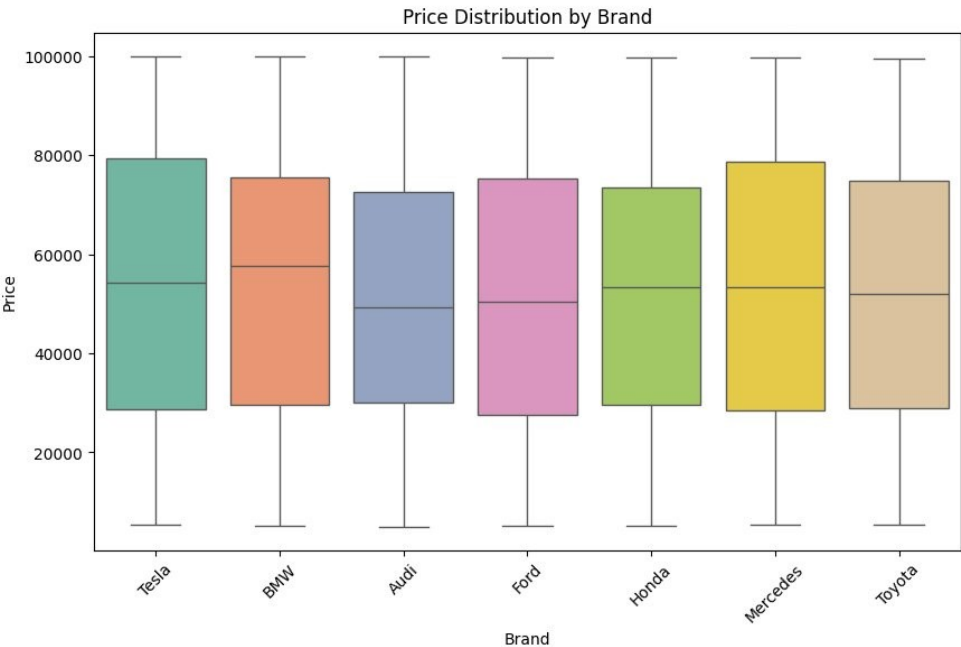


Figure 2: Price Distribution by Brand — Boxplot

Key Insights

1	BMW and Mercedes-Benz show slightly higher median prices compared to other brands, consistent with their established premium market positioning.
2	Audi exhibits a wider spread toward the lower price end, reflecting a portfolio that spans both entry-level and full luxury models.
3	Toyota and Honda remain concentrated in the mid-range price band, demonstrating consistent and predictable pricing strategies.
4	Ford displays one of the broader price ranges in the dataset, capturing both budget-conscious and higher-end vehicle segments.
5	Tesla spans a wide price range, highlighting its dual presence across mid-range and premium segments as its model portfolio has expanded.

Mileage vs. Price

This scatter analysis tests whether usage history (mileage) serves as a reliable pricing signal or whether price is determined by a broader combination of independent factors.

Code

```
sns.scatterplot(x='Mileage', y='Price', data=df, color='teal')
```

Visualisation

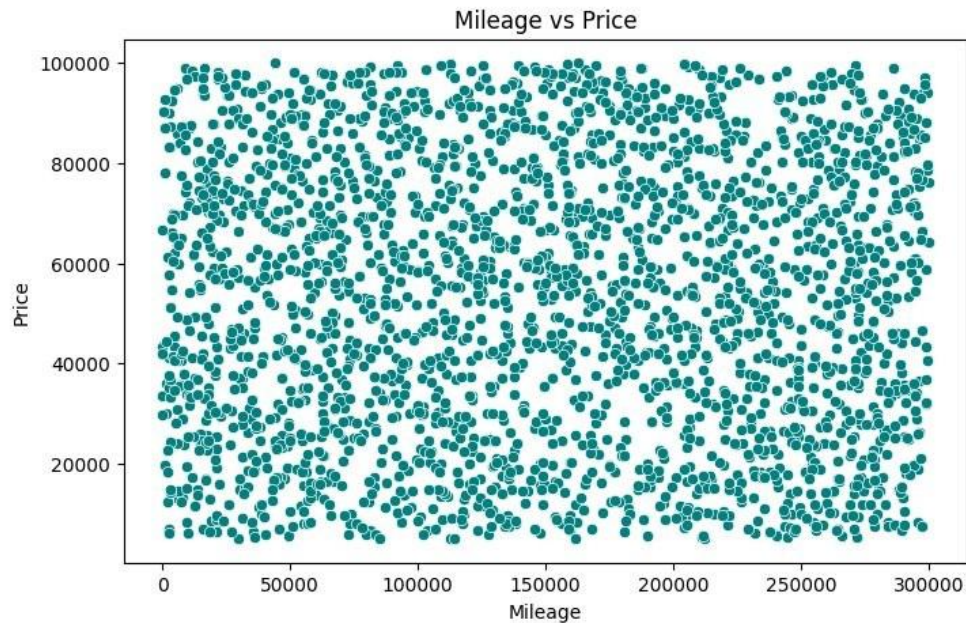


Figure 3: Mileage vs. Price — Scatter Plot

Key Insights

- 1 No clear linear relationship exists between mileage and price — the scatter confirms a weak to negligible correlation between the two variables.
- 2 High-mileage vehicles still command elevated prices, suggesting brand reputation, condition, and desirability carry more weight than mileage alone.
- 3 Low-mileage vehicles do not consistently correspond to higher prices, indicating mileage is not a primary pricing driver across this dataset.
- 4 The near-random spread of data points points to a highly diverse dataset where multiple independent variables collectively determine price.
- 5 Both budget and premium-priced vehicles appear at nearly every mileage level, confirming no single variable sufficiently explains price variation.

Average Price by Fuel Type

Comparing average prices across fuel categories reveals whether a vehicle's powertrain type commands a measurable price premium in this market.

Code

```
sns.barplot(x='Fuel Type', y='Price', hue='Fuel Type', data=df,
palette='coolwarm', legend=False)
```

Visualisation

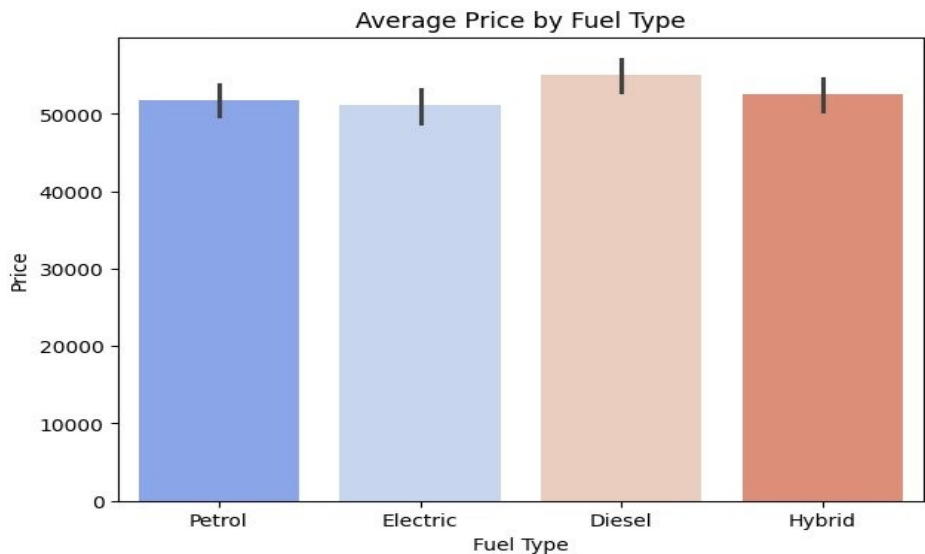


Figure 4: Average Price by Fuel Type — Bar Chart with Confidence Intervals

Key Insights

- 1 Diesel vehicles command the highest average price (~\$55,000), positioning them as the premium or heavy-duty segment within this market.
- 2 Electric vehicles average the lowest price (~\$51,000) despite the technology involved, likely reflecting government subsidies or entry-level EV models pulling the average downward.
- 3 Petrol and Hybrid vehicles both sit close to \$52,000, meaning buyers choosing between the two face no significant price difference on average.
- 4 The confidence interval for Diesel is noticeably wider than other fuel types, indicating high price variance across the segment.
- 5 Electric vehicles show the narrowest error bar, suggesting EVs are clustered around a consistent price point with comparatively low model-to-model variation.
- 6 All four fuel types fall within a ~\$4,000 range, confirming that fuel type alone is not a strong price differentiator in this dataset.

Transmission Distribution

The transmission split analysis quantifies Manual vs. Automatic representation, providing context for subsequent brand-level transmission comparisons.

Code

```
sns.countplot(x='Transmission', hue='Transmission', data=df, palette='pastel', legend=False)
```

Visualisation

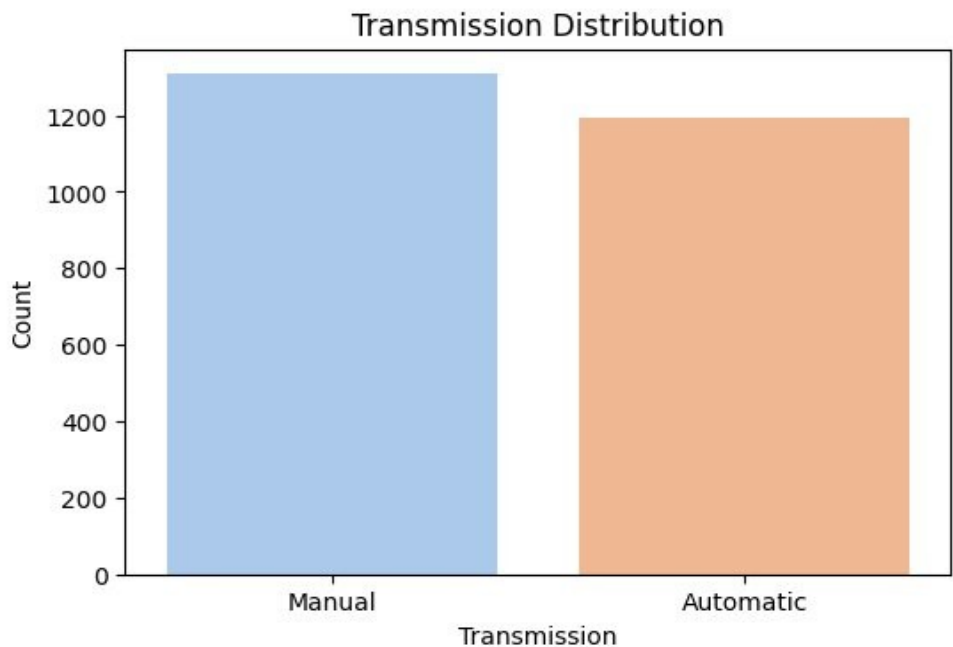


Figure 5: Transmission Distribution — Count Chart

Key Insights

- 1 Manual transmission leads the dataset with approximately 1,300 units, outnumbering Automatic vehicles by roughly 100 units.
- 2 The overall split is close to 50-50 (~52% Manual vs. ~48% Automatic), indicating the market caters effectively to both buyer preferences.
- 3 No single transmission type dominates — this dataset reflects a competitive and balanced distribution.
- 4 Automatic transmission maintains strong representation at approximately 1,200 units, consistent with growing consumer comfort toward non-manual driving.
- 5 The narrow gap of ~100 units suggests Automatic vehicles are steadily closing in, driven by urban driving conditions and shifting buyer demographics.

Correlation Matrix

The correlation matrix profiles linear relationships between all numerical features, identifying redundancy, independence clusters, and predictive leverage.

Code

```
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='magma')
```

Visualisation

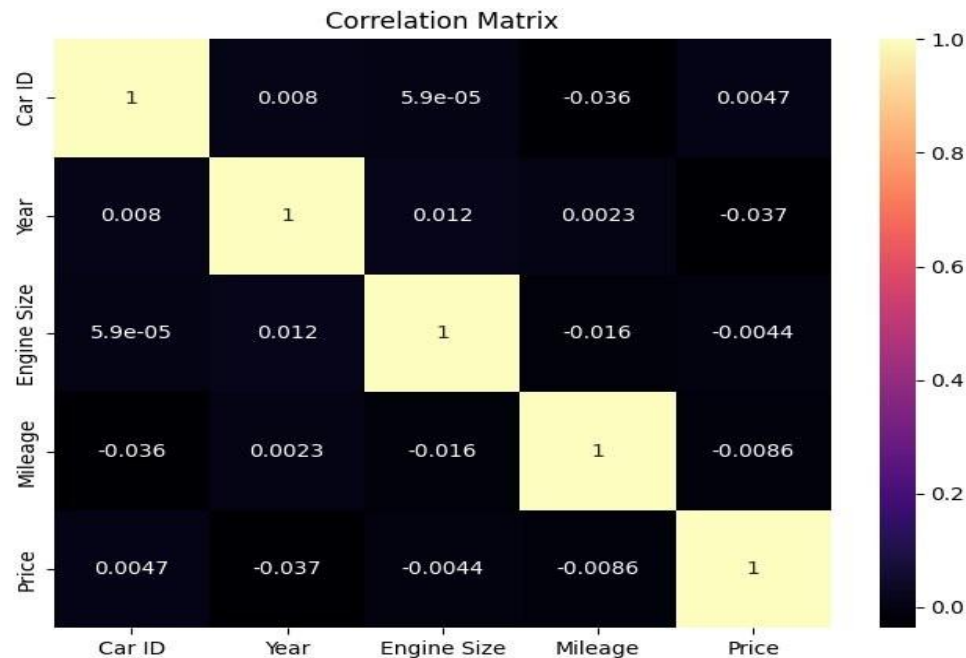


Figure 6: Correlation Matrix — Numerical Features (magma palette)

Key Insights

1	No meaningful relationship exists between any two variables — every off-diagonal correlation is extremely close to zero.
2	Price is essentially independent of all other features — correlations with Year (-0.037), Engine Size (-0.004), and Mileage (-0.009) are negligible.
3	Engine Size and Mileage show near-zero correlation (-0.016) — contrary to common intuition, larger engines are not associated with higher mileage.
4	Year has no predictive power over Price (-0.037) — newer vehicles are not meaningfully more expensive in this dataset.
5	Full feature independence confirms all variables can be safely used together in ML models without redundancy or variance inflation concerns.

Engine Size vs. Price

Testing whether engine displacement drives price variation reveals how much technical specification contributes to vehicle valuation.

Code

```
sns.scatterplot(x='Engine Size', y='Price', data=df, color='green')
```

Visualisation



Figure 7: Engine Size vs. Price — Scatter Plot

Key Insights

1	No clear correlation exists between engine size and price — the scatter is uniformly spread across all engine sizes.
2	Every engine size spans the full price range — from \$5,000 to \$100,000, vehicles of all sizes (1 to 6) appear at every price point.
3	Engine sizes 1 to 3 dominate in volume — smaller-engine vehicles are significantly more prevalent than larger-displacement alternatives.
4	Engine size 6 is the least represented — large-engine vehicles occupy a niche or premium segment with limited market volume.
5	The patternless distribution confirms engine size is an unreliable standalone predictor of price in this dataset.

Price by Condition

Examining how vehicle condition (New, Used, Like New) affects pricing tests whether condition rating is a meaningful price differentiator.

Code

```
sns.boxplot(x='Condition', y='Price', hue='Condition', data=df,
palette='cubehelix', legend=False)
```

Visualisation

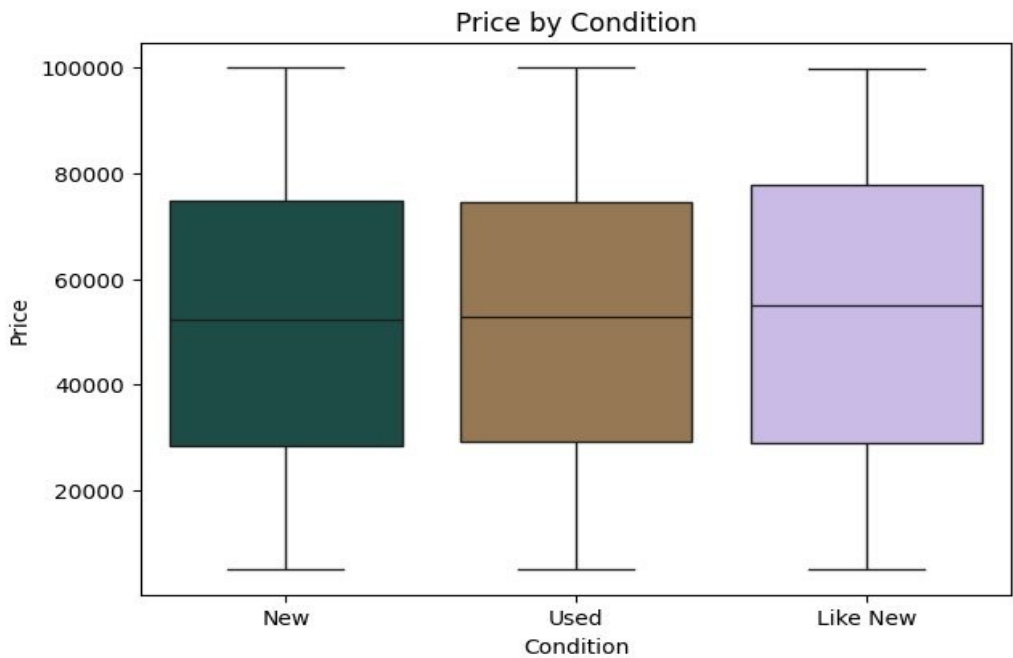


Figure 8: Price by Condition — Boxplot

Key Insights

1	All three conditions share nearly identical median prices (~\$52,000-\$55,000), indicating vehicle condition alone has minimal impact on typical selling price.
2	Like New exhibits the widest price spread — its IQR stretches the highest, reflecting greater pricing inconsistency from the subjective application of the label.
3	New vehicles show the most compact IQR — tighter and more predictable pricing consistent with standardised dealership or manufacturer-set price structures.
4	Used cars are not significantly cheaper at the median — Used median prices closely match New vehicles, pointing to high-value used inventory or premium second-hand dynamics.
5	All three conditions reach the same maximum price of ~\$100,000 — brand and features outweigh condition as value drivers in the premium segment.

Cars by Manufacturing Year

Year-by-year inventory counts reveal production and registration patterns over the 2000-2023 period, identifying peak years, troughs, and structural supply trends.

Code

```
df['Year'].value_counts().sort_index().plot(kind='line', color='red')
```

Visualisation

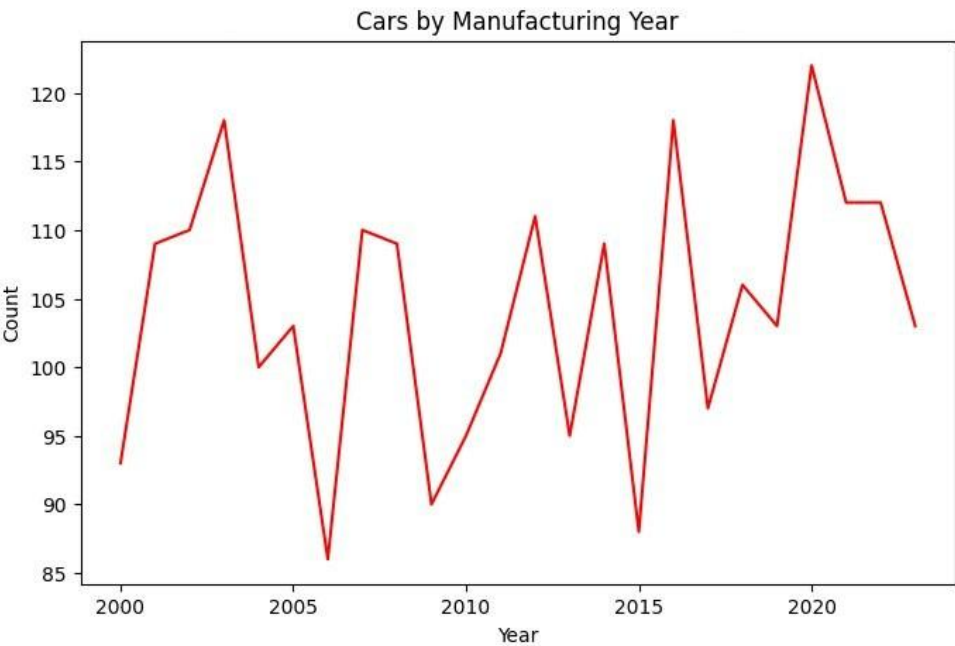


Figure 9: Cars by Manufacturing Year — Line Chart

Key Insights

1	Production volumes have been consistently volatile throughout the period — counts oscillate sharply almost every 2-3 years with no sustained growth or decline trend.
2	2020 marks the all-time peak at approximately 122 vehicles, possibly reflecting a post-pandemic inventory surge or data collection spike in recent model years.
3	2006 recorded the sharpest single-year drop — count fell to approximately 86, likely corresponding to economic slowdowns or supply chain constraints.
4	The early 2000s exhibited a rapid boom-bust cycle — a quick climb to ~118 by 2003 was followed by a steep decline by 2005.
5	Recent years (2021-2024) show a gradual tapering toward approximately 103, pointing to market saturation or recency bias in dataset coverage.

Average Mileage by Brand

Brand-level mileage comparison examines whether ownership patterns differ meaningfully across manufacturers.

Code

```
sns.barplot(x='Brand', y='Mileage', hue='Brand', data=df, palette='Blues',
            legend=False)
```

Visualisation

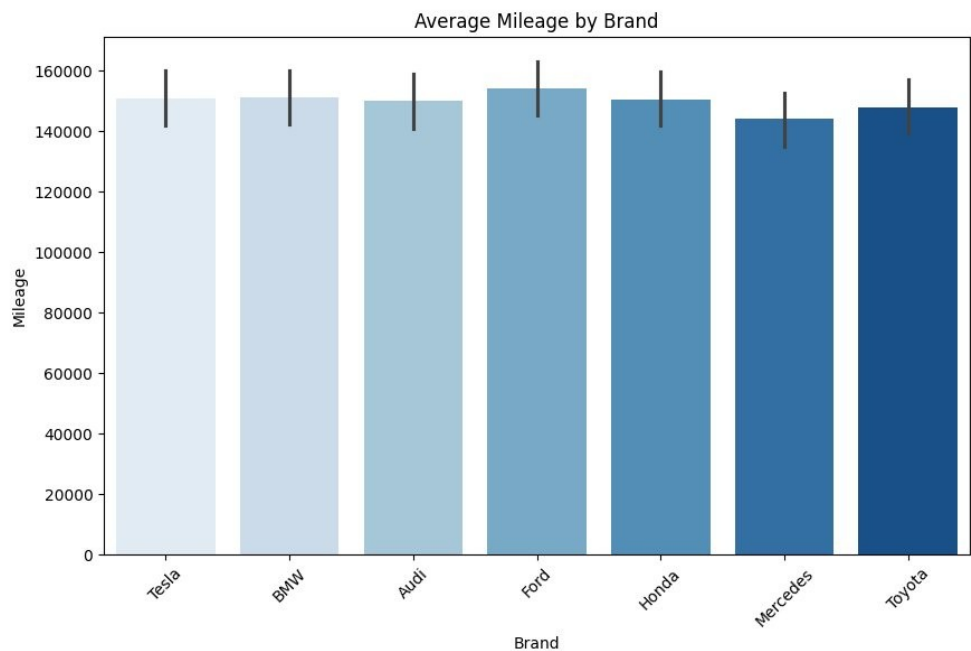


Figure 10: Average Mileage by Brand — Bar Chart with Confidence Intervals

Key Insights

- 1 Ford leads all brands in average mileage at approximately 155,000 miles, suggesting its vehicles attract high-usage buyers such as fleet operators and commercial drivers.
- 2 Mercedes-Benz records the lowest average mileage at approximately 145,000 miles, consistent with luxury ownership patterns where vehicles are driven more sparingly.
- 3 All seven brands fall within a ~10,000-mile range — brand choice has minimal impact on average mileage accumulation in this dataset.
- 4 Tesla shows the widest error bar despite being an EV-only brand, indicating high variance from low-mileage personal users to high-mileage commuters and rideshare drivers.
- 5 Ford also carries the widest confidence interval, reflecting inconsistent usage across its broad lineup from economy hatchbacks to heavy-duty trucks.

Average Price by Brand

Ranking brands by average price quantifies premium positioning and reveals whether brand identity translates to a measurable and consistent price premium.

Code

```
df.groupby('Brand')['Price'].mean().sort_values().plot(kind='barh',  
color=sns.color_palette('coolwarm', ...))
```

Visualisation

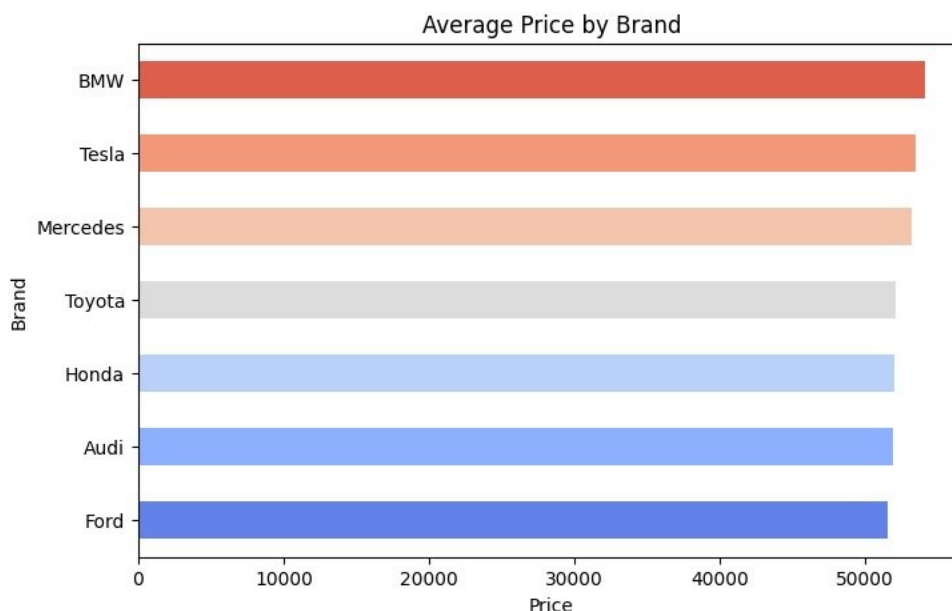


Figure 11: Average Price by Brand — Horizontal Bar Chart

Key Insights

- 1 BMW commands the highest average price at approximately \$54,000, consistent with its premium European luxury positioning.
- 2 Tesla narrowly edges out Mercedes-Benz for second place (~\$53,500 vs ~\$53,000), reflecting Tesla's successful premium brand establishment despite being a newer market entrant.
- 3 Ford records the lowest average price at approximately \$51,000, consistent with its mass-market, value-oriented brand identity.
- 4 The entire brand price range spans only ~\$3,000 — from Ford's ~\$51,000 to BMW's ~\$54,000 — confirming individual attributes are weak average price differentiators.
- 5 Audi and Honda are virtually indistinguishable in average price (~\$51,500-\$52,000), counterintuitive given Audi's premium positioning.

Average Price by Year

Tracking average price across manufacturing years reveals depreciation trends, demand cycles, and whether newer vehicles consistently command higher prices.

Code

```
df.groupby('Year')['Price'].mean().plot(color='teal')
```

Visualisation

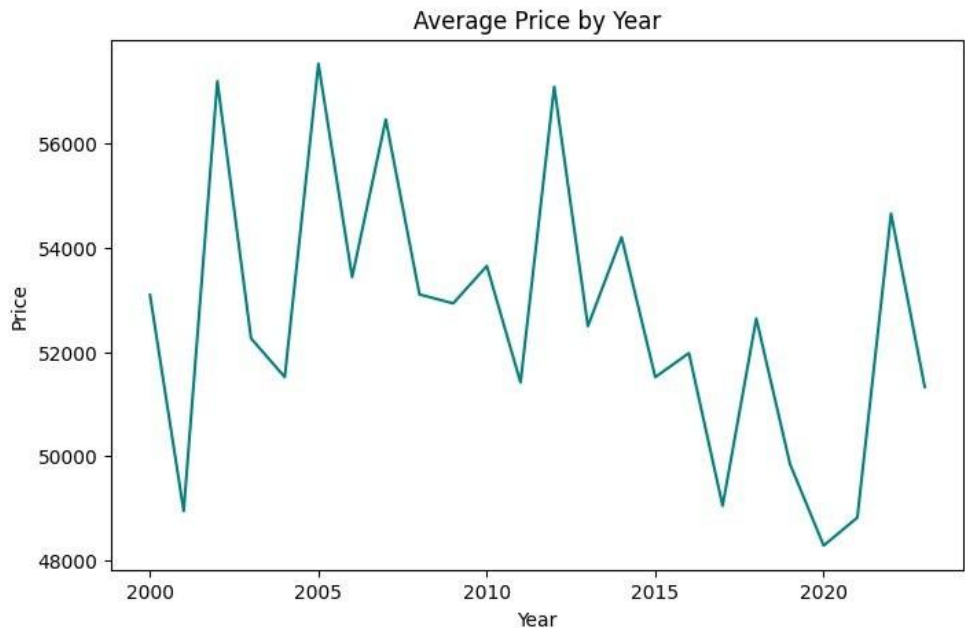


Figure 12: Average Price by Manufacturing Year — Line Chart

Key Insights

- 1 A gradual long-term downward trend is visible — the overall trajectory from ~\$53,000 in 2000 to ~\$51,500 in 2024 reflects a slow steady decline, pointing to market maturation.
- 2 2004-2005 marks the highest average price peak — prices spiked to nearly \$57,500, likely reflecting strong consumer demand or a higher concentration of premium vehicles.
- 3 2021 records the lowest average price at approximately \$48,500, possibly from post-pandemic market corrections, increased budget EV supply, or a shift in vehicle mix.
- 4 Volatility is significantly higher in the early 2000s than in recent years — sharp swings between 2000 and 2012 are more pronounced than the stable post-2015 period.
- 5 The 2022-2023 price recovery appears short-lived — a brief rebound to ~\$54,500 quickly reverses by 2024, indicating a temporary correction rather than a sustained trend.

Model Price Comparison by Brand

Model-level pricing analysis drills below brand averages to identify which specific models drive the highest value and how within-model variance compares across the dataset.

Code

```
sns.barplot(data=df, x='Model', y='Price', hue='Brand', palette='tab10')
```

Visualisation

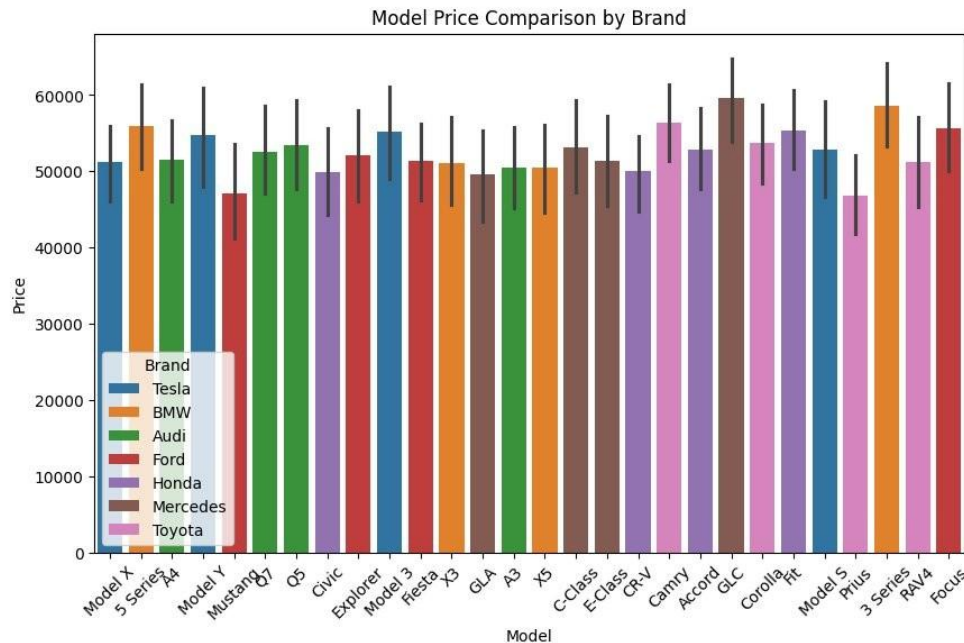


Figure 13: Model Price Comparison by Brand — Grouped Bar Chart

Key Insights

- 1 Prices are tightly clustered across all models in the \$47,000-\$60,000 range — no single model breaks dramatically away from the pack.
- 2 BMW's GLC and 3 Series command the highest model-level prices — both reach approximately \$58,000-\$59,000, consistent with BMW's brand-level lead.
- 3 Ford's Mustang sits as a clear low-end outlier at approximately \$47,000, notably below all other models despite its performance positioning.
- 4 Error bars are wide across nearly every model — tall confidence intervals indicate significant within-model price variance driven by trim level, year, and condition.
- 5 Toyota and Honda models consistently anchor the mid-price range — Camry, Corolla, CR-V, Accord, and Prius cluster tightly around \$50,000-\$55,000 with narrow error bars.

Inventory Count by Brand and Transmission

Breaking inventory down by brand and transmission type reveals which manufacturers are most Manual-heavy and where the widest supply gaps exist.

Code

```
sns.countplot(data=df, x='Brand', hue='Transmission', palette='Paired')
```

Visualisation

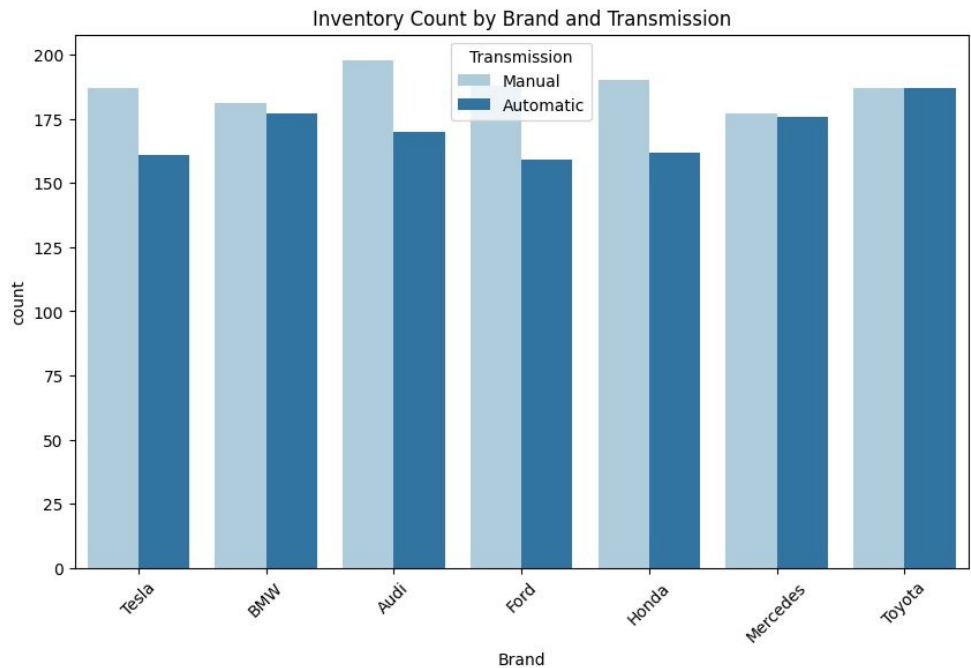


Figure 14: Inventory Count by Brand and Transmission — Grouped Bar Chart

Key Insights

- 1 Manual inventory consistently outnumbers Automatic across all seven brands — every brand shows a higher Manual count than Automatic.
- 2 Audi and Honda lead in Manual inventory — both brands reach approximately 195-200 units, making them the most Manual-heavy brands.
- 3 Toyota is the only brand where Manual and Automatic counts are nearly equal at approximately 186 units each, suggesting a deliberately balanced inventory strategy.
- 4 Ford shows the largest Manual-to-Automatic gap — Manual at ~198 vs Automatic at ~160, indicating either lower consumer demand or limited Automatic supply.
- 5 Mercedes-Benz carries the lowest overall inventory — both Manual and Automatic at approximately 175 units each, consistent with luxury low-volume positioning.

Pair Plot Analysis

The pair plot grid simultaneously visualises all bivariate relationships and individual distributions across Price, Mileage, Engine Size, and Year.

Code

```
sns.pairplot(df[['Price', 'Mileage', 'Engine Size', 'Year']])
```

Visualisation

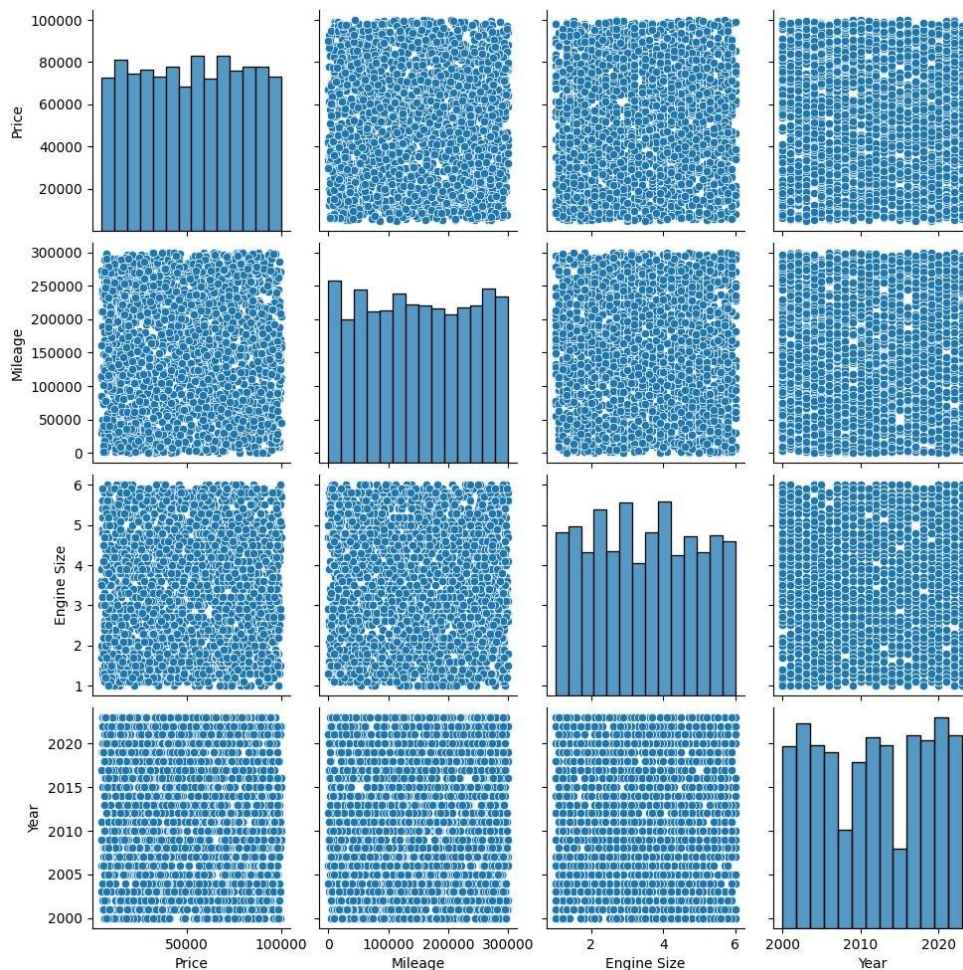


Figure 15: Pair Plot — Price, Mileage, Engine Size, Year (4x4 grid)

Key Insights

- 1 Every variable pair forms a structureless cloud — all off-diagonal scatter plots display completely random distributions, confirming no two variables share a meaningful linear relationship.
- 2 Price distribution is remarkably flat — the diagonal histogram shows near-uniform bars from \$10,000 to \$100,000, an unusual and likely curated distribution.

3

Mileage skews toward higher values — most vehicles concentrate in the 150,000-250,000 mile range, suggesting the dataset leans toward older or well-used vehicles.

4

Engine Size distribution is discrete and relatively balanced — counts spread across sizes 1 to 6, with mid-range sizes (3-5) being marginally more common.

5

Year distribution confirms the 2020 manufacturing peak — showing 2020 as the tallest bar followed by a gradual post-2021 decline.

Mileage vs. Price by Transmission Type

Segmenting the mileage-price scatter by transmission type tests whether Manual or Automatic vehicles exhibit different pricing behaviours at equivalent mileage levels.

Code

```
g = sns.FacetGrid(df, col='Transmission', height=5)
g.map_dataframe(sns.scatterplot, x='Mileage', y='Price')
```

Visualisation

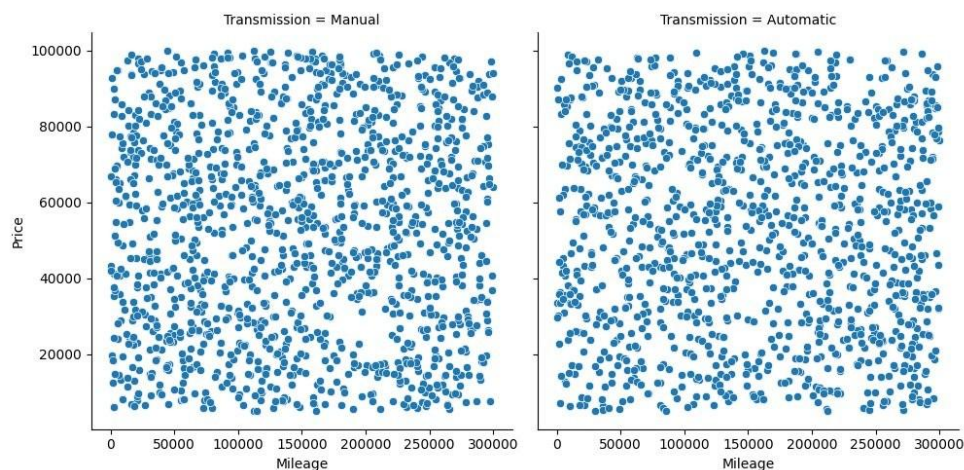


Figure 16: Mileage vs. Price by Transmission Type — Faceted Scatter Plot

Key Insights

- 1 Neither transmission type shows any mileage-price correlation — both Manual and Automatic panels display a completely random cloud pattern.
- 2 Manual transmission has a visibly higher data point density, consistent with Manual vehicles (~1,300 units) outnumbering Automatics (~1,200 units).
- 3 Both transmission types span the full mileage and price spectrum equally — from 0 to 300,000 miles and \$5,000 to \$100,000.
- 4 High-mileage vehicles command equally high prices in both categories — no visible price drop-off at 200,000+ miles in either panel.
- 5 The two panels are near-mirror images of each other — confirming transmission type contributes no additional explanatory power to the mileage-price relationship.

Brand Market Share

The brand market share pie chart quantifies how inventory is distributed across manufacturers, testing whether the dataset reflects real-world dominance or balanced sampling.

Code

```
brand_counts = df['Brand'].value_counts()
plt.pie(brand_counts, labels=brand_counts.index, autopct='%1.1f%%')
```

Visualisation

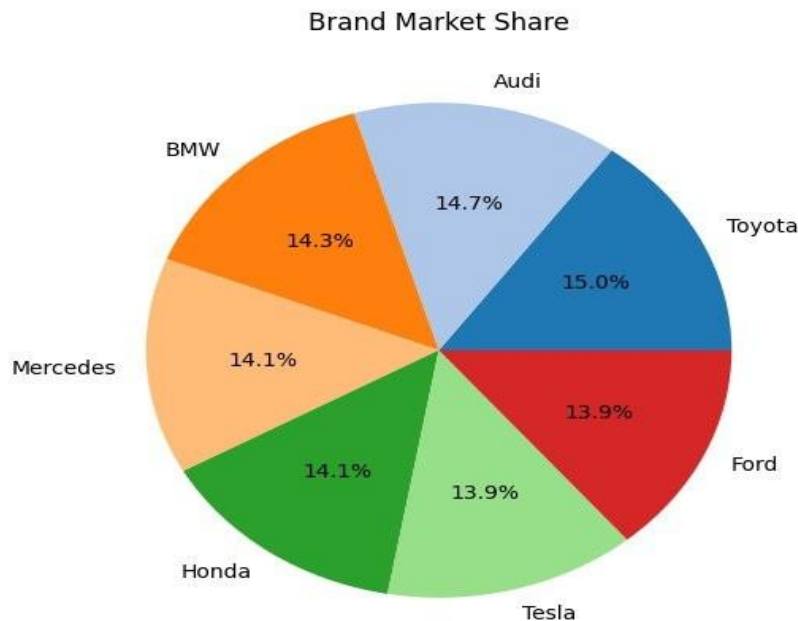


Figure 17: Brand Market Share — Pie Chart

Key Insights

1	Toyota holds the largest market share at 15.0%, yet its lead over Ford and Tesla (13.9%) is a razor-thin 1.1%, making this one of the most evenly contested brand distributions.
2	The market is remarkably balanced — every brand falls within a 1.1% range of each other (13.9%-15.0%), with no single brand dominating.
3	Ford and Tesla are tied at the bottom at 13.9%, though being last is largely meaningless given how compressed the overall distribution is.
4	Audi and BMW cluster in the mid-to-upper tier at 14.7% and 14.3%, confirming European premium brands hold ground competitively against mass-market rivals.
5	This near-uniform distribution is statistically unusual and suggests the dataset was sampled or curated to represent each brand proportionally.

Density: Mileage vs. Price

The hexbin density plot aggregates the mileage-price scatter into a binned density map, revealing whether any specific mileage-price combinations represent dominant listing patterns.

Code

```
plt.hexbin(df['Mileage'], df['Price'], gridsize=30, cmap='plasma')
```

Visualisation



Figure 18: Density — Mileage vs. Price (Hexbin, plasma palette)

Key Insights

- 1 No dominant concentration zone exists — density is spread fairly uniformly across the entire grid, confirming the absence of any dominant mileage-price combination.
- 2 High-mileage vehicles exist at every price point — even in the 250,000-300,000 mile range, vehicles appear across the full \$10,000-\$100,000 price spectrum.
- 3 Peak density hexagons (~count 9) appear without a discernible pattern, reinforcing the near-zero mileage-price correlation from the correlation matrix.
- 4 Mid-range mileage (100,000-150,000 miles) shows the subtlest concentration — slight warmth in orange and yellow tones indicates the most commonly represented mileage bracket.
- 5 Low mileage does not cluster at high prices — the even spread confirms low-mileage vehicles are not consistently priced at a premium.

Consolidated Findings & Next Steps

Overall Statistical Profile

Distribution Shape	Uniform / flat — non-normal, no dominant price band
Price Range	\$5,011 to \$99,983
Median Price	~\$53,485
Skewness	Near-zero — no meaningful directional skew
Kurtosis	Near-zero — light-tailed relative to normal distribution
Strongest Correlation	Near-zero across all numerical pairs (max ~0.037)
Top Brand by Price	BMW (~\$54,000 average)
Top Brand by Mileage	Ford (~155,000 miles average)
Brand Distribution	Near-equal at 13.9%-15.0% per brand
Transmission Split	52% Manual 48% Automatic

Consolidated Findings

1	Price structure is fundamentally uniform and non-normal — the flat distribution and near-zero skewness invalidate simple linear regression; tree-based or ensemble models are recommended.
2	All numerical features are operationally independent — near-zero correlations confirm no multicollinearity and no linear signal between any two variables.
3	Brand is the strongest categorical differentiator — but even brand-level differences span only ~\$3,000, making brand a weak standalone predictor of price.
4	Fuel type adds minimal price information — all four fuel types fall within a ~\$4,000 average price band.
5	Condition does not meaningfully separate prices — median prices for New, Used, and Like New are nearly identical.
6	Mileage has no pricing influence — confirmed consistently across scatter plots, correlation matrix, pairplot, and hexbin density analysis.
7	The dataset is almost certainly curated — near-perfectly equal brand shares, balanced condition splits, and flat price distributions are highly unlikely in a real-world sample.
8	Non-linear models are strongly recommended — Random Forest, XGBoost, or neural network architectures will substantially outperform linear approaches.

Recommended Next Steps

- **Feature Engineering** — Create interaction terms (Brand x Fuel Type, Year x Mileage) and encode categorical variables using target encoding or one-hot encoding.
 - **Baseline Modelling** — Train Linear Regression as baseline and compare against Random Forest and XGBoost to quantify non-linear improvement.
 - **Hyperparameter Tuning** — Apply GridSearchCV or Optuna to optimise tree depth, learning rate, and regularisation parameters on the best-performing model.
 - **SHAP Analysis** — Use SHAP values to interpret feature contributions at the individual prediction level.
 - **Segmented Modelling** — Train separate models per brand or segment to test whether within-group models outperform a global model.
 - **Deployment** — Export the final model as a REST API endpoint for integration into a vehicle pricing tool or dashboard.
-